🟢 Easy Level (Foundations – 40 Problems)

These are warm-up problems to build comfort with logic and syntax.

Numbers & Math

1. Swap two numbers (with and without third variable).
2. Reverse digits of a number.
3. Palindrome number check.
4. Binary → Decimal conversion (manual).
5. Decimal → Binary conversion (manual).
6. Factorial (loop + recursion).
7. Fibonacci series.
8. Prime check.
9. Sum of digits.
10. Count digits in a number.

Strings

11. Reverse a string (manual).
12. Palindrome string check.
13. Count vowels/consonants.
14. First non-repeating character.
15. Remove duplicates from string.
16. Anagram check.
17. Character frequency count.
18. Title case conversion (manual).
19. String contains only digits.
20. Longest word in sentence.

Arrays / Lists

21. Find max & min element.
22. Reverse array (manual).
23. Rotate array by k.
24. Find second largest element.
25. Remove duplicates.
26. Merge two sorted arrays.
27. Missing number in sequence.
28. Check if array is sorted.
29. Sum of elements.
30. Pair with given sum.

Patterns

31. Star pyramid.
32. Diamond pattern.
33. Hollow square.
34. Pascal's triangle.
35. Multiplication table.
36. Print odd/even numbers.
37. Print prime numbers up to n.
38. Print factors of a number.
39. Print ASCII values of characters.
40. Print Fibonacci triangle.

🟡 Medium Level (Intermediate Logic – 40 Problems)

These require deeper logic and problem-solving.

Numbers & Math

41. Armstrong number check.
42. GCD and LCM.
43. Power of a number (manual).
44. Square root approximation.
45. Generate prime factors.
46. Strong number check.
47. Automorphic number check.
48. Harshad number check.
49. Perfect number check.
50. Count trailing zeros in factorial.

Strings

51. Substring occurrence count.
52. Replace spaces with -.
53. Common characters between two strings.
54. Balanced parentheses check.
55. Remove punctuation.
56. Longest palindrome substring.
57. Reverse words in a sentence.
58. Check rotation of another string.
59. Find duplicate characters.
60. Check if two strings are isomorphic.

Arrays / Lists

61. Intersection of two lists.
62. Union of two lists.
63. Sort list manually.

64. Binary search.
65. Linear search.
66. Matrix transpose.
67. Spiral matrix printing.
68. Diagonal sum of matrix.
69. Kadane's algorithm (max subarray sum).
70. Majority element in array.

OOP

71. Bank Account class (deposit/withdraw).
72. Method overriding (Animal → Dog).
73. Method overloading (*args).
74. Shape class with area() (circle, rectangle, triangle).
75. Encapsulation with private attributes.
76. Polymorphism with multiple classes.
77. Vehicle hierarchy (Car, Bike).
78. Operator overloading (__add__).
79. Student class with grade calculation.
80. Inheritance chain (Grandparent → Parent → Child).

🔴 **Hard Level (Advanced Interview – 30+ Problems)**

These are algorithmic and data-structure heavy.

Algorithms

81. Implement quicksort.
82. Implement merge sort.
83. Implement heap sort.
84. Implement bubble sort manually.
85. Implement insertion sort manually.
86. Implement selection sort manually.
87. Find kth largest element.
88. Find median of array.
89. Find longest increasing subsequence.
90. Find longest common subsequence.

Data Structures

91. Implement stack using OOP.
92. Implement queue using OOP.
93. Reverse a linked list.
94. Detect cycle in linked list.
95. Implement doubly linked list.
96. Implement binary tree traversal (inorder, preorder, postorder).
97. Implement BFS and DFS.
98. Find height of binary tree.
99. Lowest common ancestor in binary tree.
100. Implement graph adjacency list.

Advanced Logic

101. Count words in a file.
102. Basic calculator (using OOP).
103. Generate all permutations of a string.
104. Generate all subsets of a set.
105. N-Queens problem.
106. Sudoku solver.
107. Rat in a maze problem.
108. Tower of Hanoi.
109. Knapsack problem.
110. Dijkstra's shortest path algorithm.